Alex Caramanico
John McMenimon
Carter Rasdugi

Stat 741 Final Group Project

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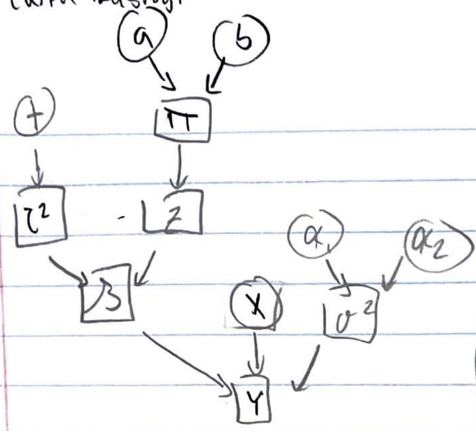
Conditional Dependence Structure



1. $y_i = x_i^T\beta_0 + \varepsilon_i$, $\varepsilon_i \sim N(0, \sigma^2\tau^2)$

$\beta_j \sim \pi\delta_0(\cdot) + (1-\pi)N(0,\tau^2)$, spike-and-slab prior

$\rightarrow \pi \sim \text{Beta}(a,b)$

$\rightarrow \tau^2 \sim IG\left(\frac{1}{2}, \frac{t^2}{2}\right)$

$\sigma^2 \sim IG(\alpha_1, \alpha_2)$

latent $z = (z_1, \dots, z_p)$

$z_j = 1$ if $\beta_j \neq 0$, otherwise $z_j = 0$

i. Derive a Gibbs Sampler for sampling posteriors

O-circled = observed or pre-set hyperparameters

□-square = random variables

$\star\ z \sim \text{Bern}(\pi)$

$p(y, \beta, z, \pi, \tau^2, \sigma^2) = p(y|\beta,\sigma^2)p(\sigma^2)p(\beta|z,\tau^2)p(z|\pi)p(\pi)p(\tau^2)$

$\star$ Need conditionals of each parameter given data & all other params

i. $p(\pi | y, \beta, z, \sigma^2, \tau^2) = p(\pi|z)$

ii. $p(\tau^2 | y, \beta, z, \pi, \sigma^2) = p(\tau^2|\beta, z)$

iii. $p(\sigma^2 | y, \beta, z, \pi, \tau^2) = p(\sigma^2|y,\beta)$

iv. $p(z|y,\beta,\pi,\tau^2,\sigma^2) = p(z|\beta,\pi,\tau^2)$

v. $p(\beta|y,z,\pi,\tau^2,\sigma^2) = p(\beta|y,z,\tau^2,\sigma^2)$

i. $p(\pi|z) \propto p(z|\pi)p(\pi) = \text{Bern}(\pi)\cdot\text{Beta}(a,b) \propto \pi^z(1-\pi)^{1-z}\cdot\pi^{a-1}(1-\pi)^{b-1}$

$= \pi^{z+a-1}(1-\pi)^{b-z}$

$\boxed{\pi|z \sim \text{Beta}(a+z, b+1-z)}$

ii. $p(\tau^2|\beta,z) \propto p(\beta|\tau^2,z)p(z)p(\tau^2)$

for $z=1$: $\propto (2\pi\sigma^2\tau^2)^{-1/2}\exp\left\{-\frac{1}{2\sigma^2\tau^2}\beta^2\right\}\left(\frac{t^2}{2}\right)^{1/2}(\tau^2)^{-\frac{1}{2}-1}\exp\left\{\frac{-\frac{t^2}{2}}{\tau^2}\right\}$

ignore parts w/o τ^2

$= (\tau^2)^{-\frac{1}{2}-1-\frac{1}{2}}\exp\left\{\frac{1}{2\sigma^2\tau^2}\beta^2 - \frac{\frac{t^2}{2}}{\tau^2}\right\}$

$= (\tau^2)^{-(1)-1}\exp\left\{-\frac{(t^2/2 + \beta^2/2\sigma^2)}{\tau^2}\right\} \rightarrow \boxed{IG\left(1, \frac{t^2}{2} + \frac{\beta^2}{2\sigma^2}\right)}$

for $z=0$: $\beta = 0 \rightarrow \boxed{IG\left(\frac{1}{2}, \frac{t^2}{2}\right)}$

iii. $p(\sigma^2|y,\beta) \propto \underbrace{p(y|\beta,\sigma^2)}_{\text{likelihood}}p(\beta)\underbrace{p(\sigma^2)}_{\text{inverse gamma}}$

$= (2\pi\sigma^2)^{-n/2}\exp\left\{\frac{-1}{2\sigma^2}\sum(y_i - \beta x_i)^2\right\}\alpha_2^{\alpha_1}(\sigma^2)^{-\alpha_1-1}\exp\left\{-\frac{\alpha_2}{\sigma^2}\right\}$

ignore parts w/o σ^2

$\propto (\sigma^2)^{-(\alpha_1+\frac{n}{2})-1}\exp\left\{\frac{-1}{\sigma^2}\left(\alpha_2 + \frac{\sum(y_i-\beta x_i)^2}{2}\right)\right\} \rightarrow \boxed{\sigma^2|y,\beta \sim \text{Gamma}\left(\alpha_1+\frac{n}{2}, \alpha_2 + \frac{\sum(y_i-\beta x_i)^2}{2}\right)}$

(matches work HW 2B result)

iv. $p(z \mid \beta, \tau^2, \theta)$

for $z=1$: $= \cancel{p(\beta \mid z=1, \tau^2, \theta)\, p(z=1 \mid \theta)}$

$\propto \boxed{(2\pi\sigma^2\tau^2)^{-1/2} \exp\{-\frac{1}{2\sigma^2\tau^2}\beta^2\} \cdot \pi}$

for $z=0$: $= \boxed{\delta_0 (1-\pi)}$

→ Markov chain won't converge w/ this, β gets stuck at 0

→ integrate over β

independent (?)

→ $\propto p(y \mid z=0, \sigma^2, \tau^2, \pi)\, p(z=0 \mid \pi)\, p(\pi)\, p(\sigma^2)\, p(\tau^2)$

$= p(y \mid z=0, \sigma^2, \tau^2, \pi)\, p(z=0 \mid \pi)$

$= \boxed{(2\pi\sigma^2)^{-n/2} \exp\{-\frac{1}{2\sigma^2}\sum y_i^2\} (1-\pi)}$

v. $p(\beta \mid y, z, \tau^2, \sigma^2)$

for $z=1$: $\propto p(y \mid \beta, z=1, \tau^2, \sigma^2)\, p(\beta \mid z=1, \tau^2)\, p(z=1)\, p(\tau^2)$

$= (2\pi\sigma^2)^{-n/2} \exp\{-\frac{1}{2\sigma^2}\sum(y_i-\beta x_i)^2\} (2\pi\sigma^2\tau^2)^{-1/2} \exp\{-\frac{1}{2\sigma^2\tau^2}\beta^2\}$

ignore terms w/o β

$\propto \exp\{-\frac{1}{2\sigma^2}\sum(y_i-\beta x_i)^2\} \exp\{-\frac{1}{2\sigma^2\tau^2}\beta^2\}$

$= \exp\{\frac{-1}{2\sigma^2}\sum(y_i-\beta x_i)^2 - \frac{1}{2\sigma^2\tau^2}\beta^2\}$

$= \exp\{\frac{-1}{2\sigma^2}[\sum(y_i-\beta x_i)^2 + \frac{1}{\tau^2}\beta^2]\}$

expand

$= \exp\{\frac{-1}{2\sigma^2}[\sum y_i^2 - 2\beta\sum y_i x_i + \beta^2\sum x_i^2 + \frac{1}{\tau^2}\beta^2]\}$

complete the square, ignore terms w/o β

$\propto \exp\{\frac{-1}{2\sigma^2}[\beta^2(\sum x_i^2 + \frac{1}{\tau^2}) - 2\beta\sum y_i x_i]\}$

call this q

$= \exp\{\frac{-(\sum x_i^2 + \frac{1}{\tau^2})}{2\sigma^2}[\beta^2 - 2\beta \boxed{\frac{\sum y_i x_i}{(\sum x_i^2 + \frac{1}{\tau^2})}}]\}$ expand... "constants" w/o β

$= \exp\{\frac{-(\sum x_i^2 + \frac{1}{\tau^2})}{2\sigma^2}[\beta - q]^2\}$ → $\boxed{N\left(\frac{\sum y_i x_i}{\sum x_i^2 + \frac{1}{\tau^2}}, \frac{\sigma^2}{\sum x_i^2 + 1/\tau^2}\right)}$

for $z=0$: $\beta \mid y, z, \tau^2, \sigma^2 \sim \boxed{\delta_0}$

Gibbs sampler would use these full conditionals sequentially to get at full posterior